$$\left(I_n - \frac{\mathbf{v}\mathbf{a}^T}{\mathbf{v}^T\mathbf{a}}\right)\cdot\mathbf{x} - \frac{a_{n+1}}{\mathbf{v}^T\mathbf{a}}\mathbf{v} \qquad \left(I_n - 2\frac{\mathbf{v}\mathbf{a}^T}{\mathbf{v}^T\mathbf{a}}\right)\cdot\mathbf{x} - 2\frac{a_{n+1}}{\mathbf{v}^T\mathbf{a}}\mathbf{v}$$

$$\Big[\cos(\theta)I_3 + \sin(\theta)[\mathbf{v}\times\_] + (1-\cos(\theta))\mathbf{v}\otimes\mathbf{v}\Big]\mathbf{x} \quad \text{where} \quad |\mathbf{v}| = 1$$

$$\begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

| $\widehat{D}$ | $D$ | $T$ | curve $C$ |
|---|---|---|---|
| $\widehat{D} = 0$ | $D > 0$ | - | A point |
| | $D = 0$ | - | Two parallel lines, a double line, or empty set |
| | $D < 0$ | - | Two lines |
| $\widehat{D} \neq 0$ | $D > 0$ | $\widehat{D}T < 0$ | An ellipse |
| | $D > 0$ | $\widehat{D}T > 0$ | The empty set |
| | $D = 0$ | - | A parabola |
| | $D < 0$ | - | A hyperbola |

| equation | name |
|---|---|
| $x^2 + y^2 + 1 = 0$ | imaginary ellipse |
| $x^2 - y^2 - 1 = 0$ | hyperbola |
| $x^2 + y^2 - 1 = 0$ | ellipse |
| $x^2 + y^2 = 0$ | two complex lines |
| $x^2 - y^2 = 0$ | two real lines |
| $x^2 + 1 = 0$ | two complex lines |
| $x^2 - 1 = 0$ | two real lines |
| $x^2 = 0$ | a real double-line |
| $x^2 - y = 0$ | parabola |

| equation | name |
|---|---|
| $x^2 + y^2 + z^2 - 1 = 0$ | ellipsoid |
| $x^2 + y^2 - z^2 - 1 = 0$ | hyperboloid of one sheet |
| $x^2 - y^2 - z^2 - 1 = 0$ | hyperboloid of two sheets |
| $x^2 + y^2 + z^2 + 1 = 0$ | imaginary ellipsoid |
| $x^2 + y^2 + z^2 = 0$ | imaginary cone |
| $x^2 + y^2 - z^2 = 0$ | (real, elliptic) cone |
| $x^2 + y^2 + 1 = 0$ | cylinder on imaginary ellipse |
| $x^2 - y^2 - 1 = 0$ | cylinder on hyperbola |
| $x^2 + y^2 - 1 = 0$ | cylinder on ellipse |
| $x^2 + y^2 = 0$ | cylinder on two complex lines |
| $x^2 - y^2 = 0$ | cylinder on two real lines |
| $x^2 + 1 = 0$ | two complex planes |
| $x^2 - 1 = 0$ | two real planes |
| $x^2 = 0$ | a double plane |
| $x^2 + y^2 - z = 0$ | elliptic paraboloid (EP) |
| $x^2 - y^2 - z = 0$ | hyperbolic paraboloid (HP) |
| $x^2 - y = 0$ | cylinder on parabola |